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Problematic digital use in adolescents with ADHD: impulsivity, obsessive-compulsive symptoms, and perceived cognitive flexibility

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Abstract

Background Adolescents with attention-deficit/hyperactivity disorder (ADHD) are at risk for problematic digital use (PDU), which is better characterised by loss of control and functional impairment than by screen time alone. The present study investigated problematic internet, social media, and gaming use among adolescents with ADHD, focusing on impulsivity, obsessive-compulsive (OC) symptoms, and perceived cognitive flexibility.

Methods This cross-sectional case-control study included 132 adolescents aged 12–18 years (69 treatment-naive adolescents with ADHD and 63 healthy controls) recruited from a tertiary child and adolescent psychiatry outpatient clinic. Participants completed self-report measures of problematic digital use, impulsivity, obsessive-compulsive symptoms, and perceived cognitive flexibility. ADHD diagnoses were confirmed using the Schedule for Affective Disorders and Schizophrenia for School-Age Children–Present and Lifetime Version (K-SADS-PL). Adjusted binary logistic regression models included participant age and biological sex. The primary within-ADHD regression and moderation models included the Conners ADHD Index.

Results Problematic internet use and probable problematic social media use were more frequent in the ADHD group and were associated with higher odds of ADHD group membership after adjustment for age and biological sex. Within the ADHD group, impulsivity and obsessive-compulsive symptom severity were independently associated with problematic internet-use severity. For gaming symptoms, impulsivity remained a significant independent correlate, whereas the OCI-R main effect was not statistically significant. A preliminary OCI-R × CFS interaction was observed for gaming symptoms after adjustment for parent-rated ADHD symptom severity (B = 0.024, SE = 0.008, 95% CI [0.007, 0.040], $p = .004$; $\Delta R^2 = 0.087$).

Conclusions Adolescents with ADHD showed higher frequencies of problematic internet use and probable problematic social media use than healthy controls. Impulsivity was a consistent dimensional correlate of problematic digital use. The interaction involving perceived cognitive flexibility should be considered preliminary and hypothesis-generating and requires replication in larger independent samples.

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Keywords ADHD, Problematic digital use, Internet gaming disorder, Impulsivity, Obsessive-compulsive symptoms, Perceived cognitive flexibility, Adolescents

Background

Digital media are integrated into adolescents' daily lives, and most use is functional. However, a minority develop patterns characterised by loss of control, unsuccessful efforts to reduce use, persistence despite negative consequences, and impairment in everyday functioning. The umbrella term problematic digital use encompasses related but non-equivalent manifestations, including problematic internet use, problematic social-media use, and gaming-disorder symptoms [1–3]. Because screen time alone does not distinguish normative engagement from clinically meaningful difficulty, symptom-based indicators of impaired control and functional consequences are particularly important [2, 4–6].

Adolescents with attention-deficit/hyperactivity disorder (ADHD) may be particularly vulnerable to problematic digital use. ADHD is characterised by developmentally inappropriate inattention and/or hyperactivity–impulsivity and is commonly associated with difficulties in self-regulation, inhibitory control, reward processing, and executive functioning [7, 8]. Digital environments frequently provide immediate feedback, variable rewards, continuous content, social reinforcement, and easy access. These features may interact with ADHD-related reward sensitivity, impulsivity, and difficulties delaying gratification or disengaging from rewarding activities [9–11].

Consistent with this rationale, children and adolescents with ADHD show higher levels of problematic digital-media use than typically developing controls [11]. Meta-analytic evidence also indicates a substantial association between ADHD symptoms and gaming-related difficulties [12]. Longitudinal evidence suggests reciprocal temporal associations between ADHD symptoms and digital-media use. Baseline ADHD symptoms have been prospectively associated with later high or problematic digital-media use, whereas baseline digital-media use has also been prospectively associated with later ADHD symptom levels [4]. These associations have been observed more consistently for problematic-use indicators than for screen-time measures, although their magnitude and direction vary across studies [4].

Two transdiagnostic constructs relevant to addiction-like behaviours are impulsivity and compulsivity. Impulsivity is defined as rapid responding to obtain reward without sufficient deliberation, whereas compulsivity is defined as repetitive, inflexible, or perseverative behaviour that continues even in the face of negative consequences [13]. Impulsivity is a core clinical feature of ADHD and may amplify sensitivity to digital environments through reward-seeking behaviour, poor inhibitory control, and difficulty delaying gratification. OC symptoms (here defined as a dimensional measure of OC symptom burden rather than a diagnosis of obsessive-compulsive disorder, OCD) may contribute via repetitive checking, difficulty terminating sessions, ordering or completion tendencies, and persistence in platform-specific goals [14–16]. More recent meta-analytic work supports the view that impulsivity is a consistent correlate of gaming disorder symptomatology [17].

A useful framework for understanding how person-level vulnerabilities interact with digital cues to produce problematic use through affective, cognitive, and executive processes is the Interaction of Person–Affect–Cognition–Execution (I-PACE) model [18]. Within this model, impulsivity may be more closely related to reward-driven initiation and escalation, whereas compulsivity-related features may be more closely related to recurrent use, checking, completion, and difficulty disengaging [18, 19]. However, to date, there has been little research on how comparable or dissimilar internet, social media, and gaming indicators are in clinically assessed ADHD populations.

Perceived cognitive flexibility and ability-based cognitive flexibility are related but non-equivalent constructs. Performance-based tasks assess observable set-shifting or task-switching under standardised conditions, whereas the Cognitive Flexibility Scale assesses individuals' subjective appraisal of their ability to generate alternatives and adapt their responses in everyday situations [20–23]. The CFS was selected because the present study concerned perceived adaptability as a person-level factor potentially relevant to digital engagement, rather than objective executive-control performance. This distinction may be particularly relevant in adolescents with ADHD, whose difficulties with self-regulation and disengagement occur in everyday, highly reinforcing contexts. Accordingly, CFS scores were interpreted exclusively as indicators of perceived cognitive flexibility.

Within the I-PACE framework, perceived cognitive flexibility may shape how OC symptom burden is expressed in digital environments [18]. Perceived adaptability was not assumed to be uniformly protective. Gaming environments require strategic adjustment while also providing structured goals, progression systems, repeated challenges, and completion contingencies [19, 24]. Among adolescents with greater OC symptom burden, perceiving oneself as able to generate alternative strategies may therefore be associated with continued optimisation or completion-oriented engagement rather

than disengagement. Accordingly, the hypothesis-driven moderation analysis focused on whether perceived cognitive flexibility modified the association between OC symptom burden and gaming symptom severity. Nevertheless, analogous interactions for problematic internet use and problematic social-media use were examined as parallel exploratory analyses to assess whether a similar pattern was present across other digital-use domains. This proposed pathway is context-dependent and hypothesis-generating and should not be interpreted as an established mechanism or as a reflection of objective executive functioning.

The present investigation was motivated by three gaps. First, few studies have examined ADHD in relation to multiple problematic digital-use indicators in a clinically assessed, treatment-naive adolescent group. Second, the separate contributions of impulsivity and OC symptom severity within ADHD have not been systematically examined while controlling for parent-rated ADHD symptom severity. Third, although there is a need to delineate perceived flexibility from performance-based executive function, perceived cognitive flexibility has rarely been investigated as a moderator of the association between OC symptoms and gaming symptoms.

Three hypotheses were tested. First, compared with healthy controls, adolescents with ADHD would be more likely to show problematic internet use, problematic social media use, and probable internet gaming disorder. Second, robust associations would be observed between impulsivity and OC symptom severity and problematic digital-use indicators within the ADHD group, which would remain significant after entering parent-rated ADHD symptom severity into the model. Third, perceived cognitive flexibility would moderate the association between OC symptom severity and gaming symptoms, such that OC symptoms would be more strongly linked with gaming severity at higher levels of perceived flexibility. Given the cross-sectional design, all results were considered as associations rather than causal processes.

Methods

Study design and participants

This cross-sectional case-control study was conducted in a tertiary child and adolescent psychiatry outpatient clinic. Both ADHD and healthy control participants aged 12–18 years were recruited by convenience sampling from the same clinic. No individual matching, frequency matching, or stratification was performed between the groups; group characteristics were compared statistically. The final sample consisted of 132 adolescents (69 treatment-naive adolescents with ADHD and 63 healthy controls).

The ADHD group was initially screened with 100 adolescents. After the structured diagnostic interview, 30 were excluded because of concurrent psychiatric diagnoses and one was excluded because of incomplete questionnaires, resulting in a final ADHD sample of 69. The control group initially included 70 adolescents, of whom 7 were excluded because of incomplete questionnaires, resulting in a final control sample of 63.

At study entry, all participants underwent the Schedule for Affective Disorders and Schizophrenia for School-Age Children–Present and Lifetime Version (K-SADS-PL) [25, 26], administered by experienced child and adolescent psychiatrists. In the ADHD group, the interview was used to confirm a DSM-5 diagnosis of ADHD and to assess current psychiatric comorbidity. Inclusion criteria for the ADHD group were: (1) age 12–18 years; (2) a DSM-5 diagnosis of ADHD confirmed using the K-SADS-PL; (3) treatment-naive status, defined as no previous pharmacological treatment for ADHD; and (4) absence of any current comorbid psychiatric disorder. Accordingly, adolescents meeting criteria for a current depressive, anxiety, obsessive-compulsive and related, trauma-related, eating, tic, disruptive-behaviour, or other psychiatric disorder were excluded. The ADHD group comprised 31 adolescents with the predominantly inattentive presentation and 38 with the combined presentation.

Inclusion criteria for the healthy control group were age 12–18 years and absence of ADHD or any current or lifetime psychiatric disorder according to the K-SADS-PL. Thus, an adolescent meeting criteria for current major depressive disorder or another current psychiatric disorder was not included in either group; any lifetime psychiatric disorder additionally precluded inclusion in the healthy control group.

Additional exclusion criteria for both groups were intellectual disability, autism spectrum disorder, bipolar disorder, psychotic disorder, substance-use disorder, use of any psychotropic medication within the previous four weeks, and chronic neurological or other medical conditions that could affect neurocognitive or behavioural outcomes. The healthy control group consisted of adolescents presenting to the same outpatient clinic for routine evaluations, such as health reports for sports or school, or for mild and transient developmental concerns, such as minor peer conflicts or mild academic stress.

The K-SADS-PL comprises an introductory interview, a screening interview, and diagnosis-specific supplements assessing current and lifetime psychopathology across major diagnostic domains, including mood, anxiety, psychotic, attention-deficit/hyperactivity and disruptive-behaviour, substance-use, eating, and tic disorders [25, 26]. Information was obtained separately from the adolescent and a parent or caregiver and integrated by the

interviewing clinician. The introductory section also elicited relevant developmental and clinical history and contextual information regarding functioning at school and at home. The interview was used for diagnostic assessment and was not treated as a separate quantitative measure of social functioning.

Measures

All measures, unless otherwise specified, were self-reported.

Sociodemographic characteristics. The sociodemographic variables included in the present analyses were participant age, biological sex, maternal age, and paternal age. These variables were recorded using a researcher-prepared information form. Birthplace, family economic condition, parental educational attainment, and broader family-level characteristics were not available for analysis.

Problematic internet use was assessed using the Young Internet Addiction Test–Short Form (YIAT-SF) [27, 28]. Scores > 30 identified the problematic/high-risk category, and scores ≥ 37 defined a higher-severity range for descriptive and sensitivity reporting. These cutoffs were used as potential risk indicators, not as clinical diagnoses.

Gaming symptoms were measured by the Internet Gaming Disorder Scale–Short Form (IGDS9-SF) [29, 30]. A total score ≥ 32 indicated probable internet gaming disorder. Problematic social media use was measured by the Social Media Disorder Scale (SMDS) [31, 32]. The presence of ≥ 5 symptoms suggested probable problematic social media use. Scale-based categories were used as probable indicators and not as diagnostic categories.

Impulsivity was evaluated using the Barratt Impulsiveness Scale-11 Short Form (BIS-11-SF) [33–35], which yields attentional, motor, non-planning, and total scores. The severity of OC symptoms was measured with the Obsessive-Compulsive Inventory–Revised (OCI-R) [36, 37]. OCI-R total scores were interpreted as dimensional indicators of self-reported OC symptom burden and not as evidence of a clinical diagnosis of OCD.

The Cognitive Flexibility Scale (CFS) [20, 38] was used to measure perceived cognitive flexibility. The CFS is a self-report measure of perceived flexibility and perspective-taking. It does not assess performance-based executive function, neurocognitive set-shifting, or task-switching capacity. Findings involving the CFS were therefore interpreted as relationships with self-perceived flexibility, not with objective cognitive control.

The severity of parent-reported ADHD symptoms was measured with the Conners Parent Rating Scale–Revised Short Form [39, 40]. The Conners ADHD Index was included as an a priori covariate in the primary within-ADHD regression and moderation models.

Sample-specific internal consistency coefficients could not be calculated because the retained analytic dataset contained computed total and subscale scores but not individual item-level responses. Reliability information from the corresponding Turkish validation studies is therefore cited in the descriptions of the measures.

Statistical analysis

Analyses were conducted using IBM SPSS Statistics and jamovi. Continuous variables are presented as means and standard deviations and were compared between groups using Welch's independent-samples t tests. Categorical variables, including biological sex and the categorical digital-use indicators, were expressed as frequencies and percentages and compared using Pearson's chi-square tests. For probable internet gaming disorder, a two-sided Fisher's exact test was used because one observed cell count was small. The descriptive comparisons reported in Table 1 were unadjusted; no ANCOVA was performed.

Multivariable binary logistic regression models estimated odds ratios (ORs) and 95% confidence intervals (CIs) for ADHD group membership associated with each problematic digital-use indicator. Participant age and biological sex were included as covariates. Maternal and paternal age were examined descriptively but were not included in the adjusted models. No measures of family functioning, parental monitoring, or parenting practices were available for inclusion as covariates. Pearson product–moment correlations between CFS, OCI-R, and BIS-11-SF total scores and the three continuous digital-use outcomes were calculated in the full sample and separately within the ADHD and healthy control groups. The Benjamini–Hochberg false discovery rate correction was applied separately to the nine total-score correlation tests within each analysis set, using pairwise-complete observations. A descriptive Pearson correlation matrix for the principal continuous study variables is presented in Supplementary Table S1.

Within the ADHD group, ordinary least-squares multiple linear regression models examined BIS-11-SF and OCI-R total scores as simultaneous correlates of YIAT-SF and IGDS9-SF scores. The primary models included the Conners ADHD Index as a covariate; corresponding models without the Conners ADHD Index were retained as unadjusted secondary analyses. Moderation analyses were conducted using observed-variable linear interaction models corresponding to a simple moderation model [41] and estimated by maximum likelihood, with coefficient inference based on two-sided Wald z tests and model-based 95% confidence intervals. The primary hypothesis-driven IGDS9-SF model included the Conners ADHD Index; the corresponding unadjusted IGDS9-SF model was secondary, and parallel unadjusted YIAT-SF and SMDS models were exploratory. OCI-R and

Table 1 Demographic and symptom-scale characteristics by diagnostic group

Variable	Healthy control (*n* = 63)	ADHD (*n* = 69)	*p*
Age, years	14.67 ± 1.71	13.91 ± 1.85	0.016*
Biological sex, n (%)			0.491
Male	30 (47.6)	37 (53.6)	
Female	33 (52.4)	32 (46.4)	
Maternal age, years	43.89 ± 6.43	40.54 ± 5.91	0.002**
Paternal age, years	46.95 ± 6.51	44.41 ± 5.65	0.019*
Conners ADHD Index†	10.38 ± 6.41	26.46 ± 6.83	< 0.001***
CFS awareness	13.59 ± 2.37	12.77 ± 5.07	0.230
CFS willingness	15.76 ± 4.51	15.00 ± 4.88	0.354
CFS self-efficacy	22.83 ± 4.18	20.45 ± 5.18	0.004**
CFS total	52.17 ± 7.91	46.88 ± 9.81	< 0.001***
OCI-R total†	22.95 ± 13.96	22.04 ± 12.01	0.690
BIS-11-SF non-planning	9.79 ± 3.29	12.67 ± 3.25	< 0.001***
BIS-11-SF motor	8.75 ± 2.89	10.45 ± 3.55	0.004**
BIS-11-SF attentional	8.68 ± 2.89	11.87 ± 3.47	< 0.001***
BIS-11-SF total	27.24 ± 6.99	34.71 ± 8.07	< 0.001***

Continuous variables are presented as mean ± standard deviation and were compared using Welch's independent-samples t tests. Biological sex is presented as frequency and percentage and was compared using Pearson's chi-square test

Conners ADHD Index ADHD Index of the Conners Parent Rating Scale–Revised Short Form, *CFS* Cognitive Flexibility Scale, *OCI-R* Obsessive-Compulsive Inventory–Revised, *BIS-11-SF* Barratt Impulsiveness Scale–11 Short Form. † Data were available for 63 healthy controls and 68 adolescents with ADHD. * $p < .05$; ** $p < .01$; *** $p < .001$

CFS scores were mean-centred before calculation of the interaction term. Conditional effects were examined at low (– 1 SD), mean, and high (+ 1 SD) CFS levels for the statistically significant unadjusted IGDS9-SF interaction. For the primary Conners-adjusted IGDS9-SF model and corresponding unadjusted secondary model, incremental variance explained by the interaction was quantified descriptively as ΔR^2 between the main-effects and interaction models; no inferential F-change test was reported. Complete-case analysis was used: the primary adjusted moderation model included 67 participants and the unadjusted models included 68. Statistical significance was defined as $p < .05$.

Results

Sample characteristics

The sociodemographic and symptom-scale characteristics of the groups are presented in Table 1. The sample included 132 adolescents (69 with ADHD and 63 healthy controls), of whom 50.8% were male. The mean age of the full sample was 14.27 ± 1.82 years. The ADHD group was younger than the healthy control group (13.91 ± 1.85 vs. 14.67 ± 1.71 years; $p = .016$). Biological sex distributions did not differ significantly: 37 adolescents with ADHD (53.6%) and 30 healthy controls (47.6%) were male ($p = .491$). Maternal age was lower in the ADHD group than in the healthy control group (40.54 ± 5.91 vs. 43.89 ± 6.43 years; $p = .002$), as was paternal age (44.41 ± 5.65 vs. 46.95 ± 6.51 years; $p = .019$). Parent-rated ADHD symptom severity was higher in the ADHD group (Conners ADHD Index: 26.46 ± 6.83 vs. 10.38 ± 6.41; $p < .001$). Adolescents with ADHD also had higher BIS-11-SF total and subscale scores. CFS awareness and willingness scores did not differ significantly between the groups ($p = .230$ and $p = .354$, respectively), whereas CFS self-efficacy and total scores were lower in the ADHD group. OCI-R total scores also did not differ significantly between the ADHD and healthy control groups (22.04 ± 12.01 vs. 22.95 ± 13.96, respectively; $p = .690$).

Problematic digital-use indicators and odds of ADHD

Problematic digital-use frequencies and ORs are shown in Table 2. Probable problematic social media use was more frequent in the ADHD group than in the healthy control group (34.8% vs. 19.0%; $p = .043$) and was associated with higher odds of ADHD group membership after adjustment for age and biological sex (adjusted OR = 2.47, 95% CI [1.09, 5.63]). Problematic/high-risk internet use was also more frequent in the ADHD group (66.7% vs. 31.7%; $p < .001$; adjusted OR = 4.80, 95% CI [2.26, 10.21]). The higher-severity YIAT-SF range showed a comparable association (50.7% vs. 20.6%; $p < .001$; adjusted OR = 4.29, 95% CI [1.95, 9.44]). Probable internet gaming disorder was more frequent in the ADHD group in the crude comparison (21.7% vs. 6.3%; two-sided Fisher's exact $p = .013$; crude OR = 4.10, 95% CI [1.28, 13.11]); an adjusted estimate was not reported because of sparse cells.

Associations of impulsivity, OC symptoms, and perceived cognitive flexibility with problematic digital-use indices

Full-sample and group-stratified correlations are presented in Table 3. In the full sample, higher CFS scores

Table 2 Problematic digital-use frequencies and odds of ADHD group membership

Variable / cutoff	Control *n* (%)	ADHD *n* (%)	*p*	Crude OR (95% CI)	Adjusted OR (95% CI)
Probable problematic social media use (SMDS ≥ 5)					
Absent	51 (81.0)	45 (65.2)	0.043*	Reference	Reference
Present	12 (19.0)	24 (34.8)		2.27 (1.02–5.05)	2.47 (1.09–5.63)
Problematic/high-risk internet use (YIAT-SF > 30)					
Normal (≤ 30)	43 (68.3)	23 (33.3)	< 0.001***	Reference	Reference
Problematic/high-risk (> 30)	20 (31.7)	46 (66.7)		4.30 (2.07–8.92)	4.80 (2.26–10.21)
Higher-severity internet-use range (YIAT-SF ≥ 37)					
Absent	50 (79.4)	34 (49.3)	< 0.001***	Reference	Reference
Present	13 (20.6)	35 (50.7)		3.96 (1.83–8.56)	4.29 (1.95–9.44)
Probable internet gaming disorder (IGDS9-SF ≥ 32)					
Absent	59 (93.7)	54 (78.3)	0.013*	Reference	—
Present	4 (6.3)	15 (21.7)		4.10 (1.28–13.11)	—

Unless otherwise indicated, p values are from Pearson's chi-square tests. The p value for probable internet gaming disorder is from a two-sided Fisher's exact test because one observed cell count was small. Crude ORs are unadjusted. Adjusted ORs were estimated using multivariable binary logistic regression including participant age and biological sex. Dashes indicate that adjusted estimates were not reported because of sparse cells

SMDS Social Media Disorder Scale, *YIAT-SF* Young Internet Addiction Test–Short Form, *IGDS9-SF* Internet Gaming Disorder Scale–Short Form. All scale-based categories indicate probable indicators, not clinical diagnoses. * $p < .05$; *** $p < .001$

Table 3 Total-score correlations with problematic digital-use outcomes in the full sample and by diagnostic group

Panel A. Full sample			
Predictor total score	**SMDS, *r* (*p*; FDR q)**	**IGDS9-SF, *r* (*p*; FDR q)**	**YIAT-SF, *r* (*p*; FDR q)**
CFS total	−0.287 (0.001; 0.001)	−0.325 (< 0.001; <0.001)	−0.330 (< 0.001; <0.001)
OCI-R total	0.334 (< 0.001; <0.001)	0.196 (0.025; 0.025)	0.327 (< 0.001; <0.001)
BIS-11-SF total	0.455 (< 0.001; <0.001)	0.419 (< 0.001; <0.001)	0.578 (< 0.001; <0.001)
Panel B. ADHD group			
Predictor total score	**SMDS, r (p; FDR q)**	**IGDS9-SF, r (p; FDR q)**	**YIAT-SF, r (p; FDR q)**
CFS total	−0.155 (0.202; 0.202)	−0.254 (0.036; 0.046)	−0.290 (0.015; 0.023)
OCI-R total	0.353 (0.003; 0.006)	0.158 (0.200; 0.202)	0.421 (< 0.001; 0.002)
BIS-11-SF total	0.396 (< 0.001; 0.002)	0.360 (0.002; 0.005)	0.517 (< 0.001; <0.001)
Panel C. Healthy control group			
Predictor total score	**SMDS, r (p; FDR q)**	**IGDS9-SF, r (p; FDR q)**	**YIAT-SF, r (p; FDR q)**
CFS total	−0.364 (0.003; 0.006)	−0.332 (0.008; 0.012)	−0.197 (0.122; 0.122)
OCI-R total	0.366 (0.003; 0.006)	0.293 (0.020; 0.022)	0.313 (0.013; 0.016)
BIS-11-SF total	0.406 (0.001; 0.004)	0.385 (0.002; 0.006)	0.473 (< 0.001; <0.001)

Values are Pearson product–moment correlation coefficients with two-sided unadjusted p values and Benjamini–Hochberg FDR-adjusted q values. FDR correction was applied separately across the nine total-score correlations within each panel. Pairwise-complete observations were used. Full-sample analyses included 132 participants, except analyses involving OCI-R ($n = 131$). ADHD-group analyses included 69 participants, except analyses involving OCI-R ($n = 68$). Healthy-control analyses included 63 participants

CFS Cognitive Flexibility Scale, *OCI-R* Obsessive-Compulsive Inventory–Revised, *BIS-11-SF* Barratt Impulsiveness Scale–11 Short Form, *SMDS* Social Media Disorder Scale, *IGDS9-SF* Internet Gaming Disorder Scale–Short Form, *YIAT-SF* Young Internet Addiction Test–Short Form

were associated with lower scores on all three problematic digital-use measures, whereas higher OCI-R and BIS-11-SF scores were associated with higher scores on all three measures. A broader descriptive correlation matrix, including the intercorrelations among the three digital-use outcomes, is provided in Supplementary Table S1.

In the ADHD group, higher YIAT-SF scores were associated with higher BIS-11-SF total scores ($r = .517$, $p < .001$, FDR q < 0.001) and higher OCI-R total scores ($r = .421$, $p < .001$, FDR q = 0.002), and with lower CFS total scores ($r = -.290$, $p = .015$, FDR q = 0.023). Higher BIS-11-SF scores were also associated with higher SMDS ($r = .396$, $p < .001$, FDR q = 0.002) and IGDS9-SF scores ($r = .360$, $p = .002$, FDR q = 0.005). Higher CFS scores were associated with lower IGDS9-SF scores ($r = -.254$, $p = .036$, FDR q = 0.046), but not with SMDS scores ($r = -.155$, $p = .202$, FDR q = 0.202). Higher OCI-R scores were associated with higher SMDS scores ($r = .353$, $p = .003$, FDR q = 0.006), whereas the association with IGDS9-SF scores was not statistically significant ($r = .158$, $p = .200$, FDR q = 0.202). Within the healthy control group, BIS-11-SF and OCI-R scores were positively associated with all three digital-use outcomes, whereas CFS scores were negatively associated with SMDS and IGDS9-SF scores but not with YIAT-SF scores (Table 3). Differences in statistical significance between groups were not interpreted as evidence that the corresponding

Table 4 Moderation of the associations between obsessive-compulsive symptoms and digital-use outcomes by perceived cognitive flexibility in the ADHD group

Panel A. Primary Conners-adjusted IGDS9-SF model					
Model	**Interaction term**	**B**	**SE**	**95% CI**	***p***
Adjusted for Conners ADHD Index	OCI-R × CFS	0.024	0.008	0.007 to 0.040	**0.004**
Panel B. Unadjusted interaction effects across digital-use outcomes					
Outcome	**Interaction term**	**B**	**SE**	**95% CI**	**p**
IGDS9-SF	OCI-R × CFS	0.022	0.009	0.005 to 0.040	**0.013**
YIAT-SF	OCI-R × CFS	0.004	0.009	−0.013 to 0.022	0.640
SMDS	OCI-R × CFS	−0.0009	0.0022	−0.0052 to 0.0035	0.696
Panel C. Conditional effects from the unadjusted secondary IGDS9-SF model					
CFS level	**Effect**	**B**	**SE**	**95% CI**	**p**
Low CFS (− 1 SD)	OCI-R slope	−0.117	0.141	−0.394 to 0.159	0.405
Mean CFS	OCI-R slope	0.101	0.103	−0.101 to 0.303	0.325
High CFS (+ 1 SD)	OCI-R slope	0.320	0.132	0.061 to 0.579	**0.015**

OCI-R Obsessive-Compulsive Inventory–Revised, *CFS* Cognitive Flexibility Scale, *IGDS9-SF* Internet Gaming Disorder Scale–Short Form, *YIAT-SF* Young Internet Addiction Test–Short Form, *SMDS* Social Media Disorder Scale, *SD* Standard deviation, *CI* Confidence interval. Values are unstandardized coefficients from observed-variable linear interaction models estimated by maximum likelihood; p values are from two-sided Wald z tests with model-based 95% CIs. OCI-R and CFS were mean-centred. The primary IGDS9-SF model adjusted for the Conners ADHD Index; the corresponding unadjusted model was secondary, and the YIAT-SF and SMDS models were exploratory. Panel C derives from the unadjusted secondary IGDS9-SF model. R^2 increased from 0.191 to 0.278 ($\Delta R^2 = 0.087$) in the primary model and from 0.080 to 0.156 ($\Delta R^2 = 0.076$) in the secondary model. The primary and unadjusted models included 67 and 68 participants, respectively. Additional decimal places are shown for SMDS. Bold values indicate $p < .05$

correlation coefficients differed, because no formal between-group comparison was performed.

ADHD-group models and moderation

In the primary Conners-adjusted multiple linear regression model for YIAT-SF scores, both BIS-11-SF (B = 0.647, SE = 0.139, 95% CI [0.368, 0.925], $p < .001$) and OCI-R scores (B = 0.353, SE = 0.092, 95% CI [0.168, 0.538], $p < .001$) were significant independent correlates; the Conners ADHD Index was not statistically significant (B = 0.218, SE = 0.165, 95% CI [− 0.111, 0.547], $p = .190$). The corresponding unadjusted secondary model showed the same pattern.

For IGDS9-SF scores, BIS-11-SF remained a significant independent correlate in the primary Conners-adjusted model (B = 0.372, SE = 0.150, 95% CI [0.072, 0.672], $p = .016$), whereas the OCI-R main effect was not statistically significant (B = 0.112, SE = 0.100, 95% CI [− 0.088, 0.311], $p = .267$). The Conners ADHD Index was positively associated with IGDS9-SF scores (B = 0.497, SE = 0.178, 95% CI [0.143, 0.852], $p = .007$). In the unadjusted secondary model, BIS-11-SF remained significant (B = 0.455, SE = 0.153, 95% CI [0.150, 0.760], $p = .004$), whereas OCI-R did not (B = 0.097, SE = 0.103, 95% CI [− 0.110, 0.304], $p = .351$).

Table 4 presents the primary adjusted and secondary unadjusted moderation analyses. In the primary model adjusted for the Conners ADHD Index, the OCI-R × CFS interaction for IGDS9-SF scores was statistically significant (B = 0.024, SE = 0.008, 95% CI [0.007, 0.040], $p = .004$). The explained variance increased from $R^2 = 0.191$ in the main-effects model to $R^2 = 0.278$ after inclusion of the interaction term ($\Delta R^2 = 0.087$).

The corresponding unadjusted secondary interaction was also statistically significant (B = 0.022, SE = 0.009, 95% CI [0.005, 0.040], $p = .013$). In this model, explained variance increased from $R^2 = 0.080$ to $R^2 = 0.156$ ($\Delta R^2 = 0.076$). Conditional-effect estimates from the unadjusted model indicated that OCI-R scores were positively associated with IGDS9-SF scores at high CFS levels (+ 1 SD; B = 0.320, SE = 0.132, 95% CI [0.061, 0.579], $p = .015$), but not at the mean CFS level (B = 0.101, SE = 0.103, 95% CI [− 0.101, 0.303], $p = .325$) or at low CFS levels (− 1 SD; B = − 0.117, SE = 0.141, 95% CI [− 0.394, 0.159], $p = .405$).

Parallel exploratory unadjusted models did not identify statistically significant OCI-R × CFS interactions for YIAT-SF scores (B = 0.004, SE = 0.009, 95% CI [− 0.013, 0.022], $p = .640$) or SMDS scores (B = − 0.0009, SE = 0.0022, 95% CI [− 0.0052, 0.0035], $p = .696$). Among the three outcomes examined, a statistically significant interaction was observed only for gaming symptom severity in the present sample; however, the interaction coefficients were not formally compared across outcomes. Given the relatively small ADHD subgroup, the IGDS9-SF interaction and ΔR^2 estimates should be considered preliminary and hypothesis-generating.

Discussion

This study examined problematic digital-use indicators in adolescents with ADHD, focusing on impulsivity, OC symptom severity, and perceived cognitive flexibility. Four findings merit emphasis. First, problematic internet use and probable problematic social media use were more frequent in the ADHD group and were associated with higher odds of ADHD group membership after adjustment for age and biological sex. Second,

probable internet gaming disorder was more frequent in the ADHD group in the crude comparison, although an adjusted binary estimate was not reported because of sparse cells. Third, within the ADHD group, impulsivity was independently associated with both problematic internet-use and gaming-symptom severity, whereas OC symptom severity showed an independent main effect for problematic internet use but not gaming symptoms. Fourth, a preliminary OCI-R × CFS interaction was observed for gaming symptoms, including after adjustment for parent-rated ADHD symptom severity.

Elevated odds of problematic digital use in ADHD

The between-group findings are consistent with previous evidence indicating elevated problematic digital-use indicators among adolescents with ADHD [11, 12]. The strongest signal concerned problematic internet use: crossing the problematic/high-risk threshold was associated with higher odds of ADHD group membership, and the more stringent higher-severity threshold yielded a comparable result. Probable problematic social media use was also associated with higher odds of ADHD group membership, although the magnitude was smaller.

This pattern is consistent with a broad but not unlimited view of digital risk in ADHD. The findings do not imply that all digital platforms are equally harmful, or that screen-time quantity alone explains the association. Loss of control, repeated unsuccessful attempts to cut down use, and impairment in daily functioning are more likely to be the clinically meaningful signal [2, 4]. This is consistent with existing PUI frameworks that view problematic online activities as related indicators under a broader umbrella rather than independent constructs [2, 3].

Gaming symptoms warrant a more nuanced interpretation. In the crude categorical comparison, probable internet gaming disorder was more frequent in adolescents with ADHD. Within the ADHD group, gaming severity was positively correlated with impulsivity, whereas its bivariate association with OCI-R score was not statistically significant. The regression and moderation results likewise suggest that OC symptom burden was not associated with gaming severity through a uniform main effect; rather, the association varied according to perceived cognitive flexibility. These findings should be interpreted cautiously given the overlap among the digital-use indicators and the relatively small ADHD subgroup.

Impulsivity and OC symptoms as clinically relevant correlates

Impulsivity was consistently associated with problematic digital use. This is consistent with transdiagnostic models that emphasise reward-seeking, poor inhibitory control, and rapid cue-reactivity in addiction-like behaviours [13]. Impulsivity is a core diagnostic feature of ADHD [9, 18], and digital environments that provide instant feedback and a variety of rewards may be particularly difficult to disengage from. The persistence of associations between impulsivity and digital-use indicators after adjusting for parent-rated ADHD symptom severity suggests that impulsivity carries clinically relevant dimensional information that is not reducible to either diagnostic status or global ADHD severity.

OC symptom severity showed an independent association with problematic internet-use severity within the ADHD group. The absence of a between-group difference in OCI-R total scores does not preclude such dimensional within-group associations, particularly because adolescents with current psychiatric comorbidity, including OCD, were excluded. For gaming symptoms, however, the OCI-R main effect was not statistically significant; the observed association varied according to perceived cognitive flexibility. These patterns are compatible with the possibility that repetitive checking, completion tendencies, ordering, and difficulty stopping may be relevant in some digital-use contexts, but they do not establish a direct compulsivity mechanism [14–16, 19].

OCI-R scores were interpreted as self-reported dimensional indicators of OC symptom burden rather than evidence of clinical OCD. This distinction is important because the OCI-R is a symptom measure, whereas current psychiatric comorbidity was assessed using the K-SADS-PL [36].

Perceived cognitive flexibility and gaming symptoms

The between-group CFS findings were not uniformly null. Adolescents with ADHD had lower CFS self-efficacy and total scores, whereas awareness and willingness scores did not differ significantly between groups. This pattern may indicate that differences in perceived cognitive flexibility are facet-specific rather than global. Because the CFS assesses subjective appraisal of adaptability rather than performance-based set-shifting, the absence of differences on individual subscales should not be interpreted as evidence of intact objective cognitive flexibility. Conversely, the present study cannot determine whether discrepancies between self-perception and objective performance contributed to these findings because no performance-based flexibility measure was administered.

The preliminary interaction finding may help generate hypotheses regarding gaming-related processes in adolescents with ADHD. The OCI-R × CFS interaction accounted for an additional 8.7% of the variance in gaming symptom severity in the primary model adjusted for parent-rated ADHD symptom severity and 7.6% in the corresponding unadjusted secondary model. Although

these estimates indicate incremental explanatory contribution beyond the main effects, they were obtained from a relatively small ADHD subgroup and may be sample-dependent. Accordingly, the interaction should not be interpreted as evidence of a stable or clinically established moderating mechanism. In the unadjusted conditional-effects analysis, higher OC symptom severity was associated with greater gaming severity only at high levels of perceived cognitive flexibility.

One possible, speculative interpretation is that adolescents who perceive themselves as highly adaptable may remain engaged in gaming environments that reward planning, goal achievement, and strategic adjustment. Among adolescents with greater OC symptom burden, this perceived adaptability may be associated with a more repetitive or completion-driven pattern of gaming engagement and greater difficulty disengaging. However, the present cross-sectional findings cannot establish such a mechanism. Self-perceptions of adaptability may have context-dependent associations [24].

The conditional pattern should not be interpreted as evidence that perceived cognitive flexibility is clinically harmful or lacks a protective role. Parallel exploratory models did not show statistically significant interactions for problematic internet-use or social-media-use scores, but the interaction coefficients across outcomes were not formally compared. Replication in larger samples using longitudinal designs, objective usage measures, and performance-based measures of cognitive flexibility is required.

Clinical implications

The findings suggest that clinical assessment of adolescents with ADHD should not be restricted to screen time. Relevant signals include loss of control, unsuccessful efforts to reduce use, functional impairment, and repetitive or completion-driven engagement. Impulsivity may help identify broad vulnerability across digital-use domains, whereas assessment of dimensional OC symptom burden may help characterise patterns with a more repetitive, rigid, or completion-driven quality. These interpretations do not establish an OCD diagnosis or a specific causal mechanism.

The present interaction finding is not sufficient to support specific treatment recommendations concerning flexibility-oriented interventions. Rather, it identifies a preliminary question for future clinical and longitudinal research regarding how perceived adaptability and OC symptom burden may jointly relate to persistent gaming engagement. Family-level assessment may also be informative, given that family functioning has been linked to PUI severity in youth [42] and that parental monitoring and digital norms were not directly assessed.

Strengths and limitations

Strengths of this study include the use of a treatment-naive sample of adolescents with ADHD, comprehensive structured diagnostic assessment, simultaneous examination of multiple digital-use indicators, group-stratified correlation analyses, and inclusion of parent-rated ADHD symptom severity in the primary within-ADHD models. Several limitations should also be noted.

First, the cross-sectional design precludes causal inference, including conclusions about whether ADHD-related vulnerabilities contribute to problematic digital use or whether problematic digital use contributes to later ADHD symptom severity. The use of self-report measures may also inflate shared method variance, and the digital-use categories represent probable indicators rather than clinical diagnoses. Second, perceived cognitive flexibility was assessed by self-report and should not be generalised to performance-based executive functioning. Third, although the incremental variance explained by the OCI-R × CFS interaction was quantified using ΔR^2, the moderation models were estimated in a relatively small ADHD subgroup; the interaction and effect-size estimates may therefore be unstable and should be regarded as hypothesis-generating. Johnson–Neyman regions were not examined. Fourth, the three digital-use outcomes were intercorrelated, indicating shared variance, although they were analysed as separate outcomes and should not be regarded as interchangeable constructs. Fifth, maternal and paternal age differed between groups and were not included as covariates, so residual family-level confounding cannot be excluded. Birthplace, family economic condition, parental educational attainment, family functioning, parental monitoring, and parenting practices were not available for analysis. Finally, sample-specific internal consistency coefficients could not be calculated because item-level responses were not retained in the analytic dataset.

Conclusions

Adolescents with ADHD showed higher frequencies of problematic internet use and probable problematic social media use than healthy controls. Within the ADHD group, impulsivity was independently associated with problematic internet-use and gaming-symptom severity, whereas OC symptom severity showed an independent association with problematic internet use but not a uniform main effect for gaming symptoms. A preliminary OCI-R × CFS interaction was observed for gaming symptom severity and accounted for incremental variance beyond the corresponding main effects. This finding requires replication in larger independent samples and should not be interpreted as evidence of a stable or clinically established moderating mechanism.

Abbreviations

ADHD	Attention-deficit/hyperactivity disorder
BIS-11-SF	Barratt Impulsiveness Scale-11 Short Form
CFS	Cognitive Flexibility Scale
CI	Confidence interval
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, 5th edition
FDR	False discovery rate
IGDS9-SF	Internet Gaming Disorder Scale–Short Form
I-PACE	Interaction of Person–Affect–Cognition–Execution
K-SADS-PL	Schedule for Affective Disorders and Schizophrenia for School-Age Children–Present and Lifetime Version
OC	Obsessive-compulsive
OCD	Obsessive-compulsive disorder
OCI-R	Obsessive-Compulsive Inventory–Revised
OR	Odds ratio
PDU	Problematic digital use
PUI	Problematic usage of the internet
SD	Standard deviation
SMDS	Social Media Disorder Scale
YIAT-SF	Young Internet Addiction Test–Short Form

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

DAV contributed to study concept and design, formal analysis, data curation, and writing the original draft and the review and editing of the manuscript. HB contributed to study concept and design and formal analysis. SG contributed to study supervision and resources. CY, EGYA, EÇ, GA, and PŞ contributed to data curation. Nİ contributed to data curation and to review of the manuscript. All authors had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. All authors read and approved the final manuscript.

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Data availability

The datasets usedand analysed during the current study are available from the corresponding author on reasonable request. Data are not publicly available because theycontain information that could compromise the privacy of the adolescent participants.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and was approved by the Clinical Research Ethics Committee of Adana City Training and Research Hospital (date: September 25, 2025; meeting no. 17; decision no. 751). Written informed consent was obtained from the parents or legal guardians of all participants, and assent was obtained from the adolescent participants.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form (including any individual details, images, or videos).

Competing interests

The authors declare no competing interests.

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